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Interfacial charge and thermal transport in diamond field-effect transistors with c-BN and Al_2O_3 gate dielectrics

Min Hyeok Kim [a], Byoung Don Kong [a,*]

[a] Pohang University of Science and Technology (POSTECH), Pohang, Gyeongsangbuk-do, 37673, Republic of Korea

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ABSTRACT

Diamond is a favorable semiconductor for high-power and high-temperature electronics; however, the electrothermal stability of diamond field-effect transistors is often limited by interfacial heat dissipation and the thermal stability of interface-induced charge. In diamond field-effect transistors that rely on interfacial carrier channels, both interfacial thermal resistance and interfacial electrostatics play critical roles in high-temperature operation. Here, we present a multiscale computational framework that connects atomistic interfacial properties to device-level electrothermal behavior. Interfacial thermal resistance is evaluated using phonon Green's function calculations with first-principles phonon spectra for crystalline diamond/cubic boron nitride interfaces and diffuse mismatch modeling for diamond/amorphous-Al_2O_3 interfaces. The calculated interfacial thermal resistance values are 5.03×10^{-10} m^2 K W^{-1} for diamond/cubic boron nitride(100) and 5.95×10^{-10} m^2 K W^{-1} for (111), nearly an order of magnitude lower than the calibrated estimate of (5.19–5.40) $\times 10^{-9}$ m^2 K W^{-1} for diamond/amorphous-Al_2O_3. First-principles electrostatic analysis incorporating lattice expansion from 300 K to 1000 K shows that polarization-induced interfacial charge at diamond/c-BN interfaces is weakly temperature dependent, varying from -1.5×10^{12} to -1.8×10^{12} cm^{-2} for C-B termination and from 2.6×10^{13} to 2.5×10^{13} cm^{-2} for C-N termination. Incorporating these interfacial parameters into calibrated electrothermal device simulations reveals that Al_2O_3-gated diamond field-effect transistors exhibit ~40% reduction in saturated drain current at elevated temperature, whereas cubic boron nitride-gated devices show only ~20% reduction with minimal threshold shift. These results demonstrate how interfacial electrostatics and thermal resistance jointly govern electrothermal performance in diamond field-effect transistors and underscore the importance of dielectric and interface selection for high-temperature diamond electronics.

1. Introduction

Diamond is an attractive semiconductor for high-power and high-temperature electronics due to its ultra-wide bandgap (~5.47 eV), high thermal conductivity (~2000 W m^{-1} K^{-1}), high carrier mobilities (up to 4500 cm^2 V^{-1} s^{-1} for electrons and 3800 cm^2 V^{-1} s^{-1} for holes), and breakdown field exceeding 10 MV cm^{-1} [1]. These intrinsic properties surpass those of established wide-bandgap semiconductors such as GaN and SiC, positioning diamond as a candidate for operation in extreme thermal and electrical environments. However, practical device implementation has been limited by the difficulty of achieving reliable bulk doping, owing to large dopant activation energies and limited solubility.

Consequently, diamond field-effect transistors (FETs) predominantly rely on interface-induced carrier channels rather than conventional bulk doping. Most devices have utilized hydrogen-terminated diamond surfaces with adsorbate-induced surface transfer doping, where strong acceptor species such as NO_2 generate a two-dimensional hole gas [2–7]. Although this approach enables high sheet carrier densities, its stability is limited by the weak binding of adsorbates and sensitivity to thermal and environmental conditions, restricting reliable operation to temperatures below approximately 200 °C [8].

To improve thermal robustness, significant efforts have focused on gate-stack engineering using high-temperature atomic layer deposition (ALD) of high-k dielectrics, most notably amorphous Al_2O_3 (a-Al_2O_3). Fixed interfacial charges at the diamond/a-Al_2O_3 interface enable adsorption-free channel formation and extend stable operation to temperatures approaching ~400 °C [9–15]. However, the fixed-charge densities achievable in a-Al_2O_3 are typically lower than those obtained through surface transfer doping, resulting in reduced sheet carrier

* Corresponding author.
E-mail address: bdkong@postech.ac.kr (B.D. Kong).

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density and limited current drive. This trade-off between thermal stability and interfacial charge density motivates exploration of alternative dielectric interfaces.

Cubic boron nitride (c-BN) has emerged as a promising candidate to address this limitation. c-BN exhibits a wide bandgap (~6.4 eV), high thermal conductivity (up to ~1600 W m^{-1} K^{-1} with isotopic enrichment) [16,17], and good lattice compatibility with diamond. The diamond/c-BN heterointerface is predicted to support polarization- and bonding-induced interfacial charge arising from heterovalent bonding and band alignment, potentially enabling stable, high-density carrier accumulation without reliance on chemically fragile surface terminations [18–22]. Advances in epitaxial c-BN growth [23–25] and the availability of large-area diamond substrates motivate systematic assessment of this heterostructure for high-temperature diamond electronics.

In diamond FETs, carrier transport occurs in a nanoscale interfacial channel where Joule heating is generated directly at the gate dielectric interface. Under such conditions, electrothermal behavior is governed not only by bulk thermal conductivities but also by interfacial properties, including interfacial thermal resistance (ITR) and the thermal stability of interfacial charge. While bulk thermal properties of diamond and common dielectrics are well established, quantitative understanding of ITR at diamond/a-Al_2O_3 and diamond/c-BN interfaces remains limited. Moreover, the temperature dependence of interfacial charge formation at diamond/c-BN interfaces has not been systematically quantified.

In this work, we present a multiscale investigation of interfacial electrostatics and thermal transport in diamond FETs employing a-Al_2O_3 and c-BN gate dielectrics. Interfacial charge formation at diamond/c-BN interfaces is quantified using first-principles electrostatic analysis incorporating lattice expansion. ITR is evaluated using phonon Green's function methods for crystalline diamond/c-BN interfaces and diffuse mismatch modeling for diamond/a-Al_2O_3 interfaces. These interfacial properties are incorporated into calibrated electrothermal device simulations to assess temperature-dependent current drive, channel heating, and threshold behavior. The results clarify how interfacial charge stability and ITR jointly govern electrothermal behavior in diamond FETs under elevated-temperature operation.

2. Interfacial charge modeling

Interfacial charge formation at the diamond/c-BN interface was evaluated by extracting the polarization-induced sheet charge density from first-principles electrostatic calculations. The interfacial sheet charge density was determined from the discontinuity of the electric field across the interface according to

$$\overline{p}(z) = \varepsilon_0 (E_{Diamond} - E_{BN}), \tag{1}$$

where E_{Diamond} and E_{BN} are the electric fields obtained from linear fits to the bulk-like regions of the macroscopic electrostatic potential on the diamond and c-BN sides of the interface, respectively, and ε_0 is the vacuum permittivity. This approach captures the intrinsic polarization charge associated with heterovalent bonding at the diamond/c-BN heterointerface [18,26]. Artificial dipole corrections and hydrogen termination were not applied in order to isolate intrinsic interfacial polarization effects.

Temperature dependence was incorporated structurally through lattice expansion of diamond up to 1000 K, rather than through explicit finite-temperature electronic effects. Temperature-dependent lattice constants were derived from the calculated ground-state lattice constant a_0 by integrating the linear thermal expansion coefficient $\alpha(T)$:

$$a(T) = a_0 \exp\left(\int_{T_0}^{T} \alpha(t) dt \right), \tag{2}$$

where T_0 is a reference temperature. The thermal expansion coefficient $\alpha(T)$ was adopted from the multi-frequency Einstein model ("fit 1") reported by Jacobson et al. [30], which accurately reproduces experimental thermal expansion data of monocrystalline diamond from 10 K to 1000 K:

$$\alpha(T) = \sum_{i=1}^{4} X_i E\left(\frac{\Theta_i}{T}\right), \tag{3}$$

with $E(x) = (x^2 e^x)/(e^x - 1)^2$. The parameters $\{X_i, \Theta_i\}$ were taken directly from the reference. Numerical evaluation yields diamond lattice constants of 3.572 Å at 300 K, 3.574 Å at 500 K, 3.576 Å at 700 K, and 3.580 Å at 1000 K. These lattice parameters were used to construct temperature-dependent diamond/c-BN interface supercells.

For each temperature and bonding configuration, the planar-averaged Kohn–Sham electrostatic potential $U(z)$ was computed and smoothed using a macroscopic averaging procedure,

$$\overline{U}(z) = \frac{1}{d} \int_{z-d/2}^{z+d/2} U(z') dz', \tag{4}$$

where the averaging window d matches the lattice periodicity along the interface-normal direction [27–29]. This procedure removes atomic-scale oscillations and yields well-defined plateau regions in the bulk portions of diamond and c-BN, enabling reliable extraction of the electric-field slopes used in Eq. (1).

3. Charge density at the diamond/c-BN interface

Using the modeling framework described above, interfacial charge densities were evaluated for diamond/c-BN (100) interfaces with C-B and C-N bonding configurations over the temperature range from 300 to 1000 K. The electrostatic potential was obtained from density functional theory calculations implemented in the Quantum ESPRESSO package [31]. The calculations employed the generalized gradient approximation with the Perdew–Burke–Ernzerhof exchange–correlation functional and optimized norm-conserving Vanderbilt pseudopotentials [32]. A plane-wave energy cutoff of 100 Ry and a 12 × 12 × 1 Monkhorst–Pack k-point grid were used to ensure convergence of the interfacial electrostatic potential.

To model pseudomorphic diamond/c-BN heterostructures, slab supercells were structurally optimized with the in-plane lattice constant constrained to that of diamond, as shown in Fig. 1(a). The optimized diamond lattice constant obtained in this work is $a_0 = 3.567$ Å, consistent with the experimentally reported value of 3.567 Å [33]. Given the relatively small lattice mismatch between diamond and c-BN (~1.6%) [34], the c-BN overlayer is expected to remain coherently strained, supporting the pseudomorphic approximation adopted in the calculations. Representative macroscopic averages of the electrostatic potential profiles for the two bonding configurations at 300 K are shown in Fig. 1(b) and (c), and the extracted interfacial sheet charge densities are summarized in Table 1.

A pronounced polarity dependence is observed. The C–N–terminated interface exhibits a large positive sheet charge density exceeding 10^{13} cm^{-2}, whereas the C–B–terminated interface yields a substantially smaller negative sheet charge density. For both bonding configurations, the magnitude of the interfacial charge shows only weak variation with temperature from 300 to 1000 K. This weak temperature dependence indicates that interfacial charge formation at the diamond/c-BN interface is governed primarily by intrinsic heterovalent bonding and the associated interface polarization, rather than by thermal lattice expansion. Consequently, the induced interfacial charge can be regarded as thermally robust over the temperature range relevant to high-power and high-temperature device operation.

While a fully C–N–terminated interface maximizes the positive interfacial charge density, it is not a strict requirement for achieving net electron accumulation. Experimental observations and theoretical

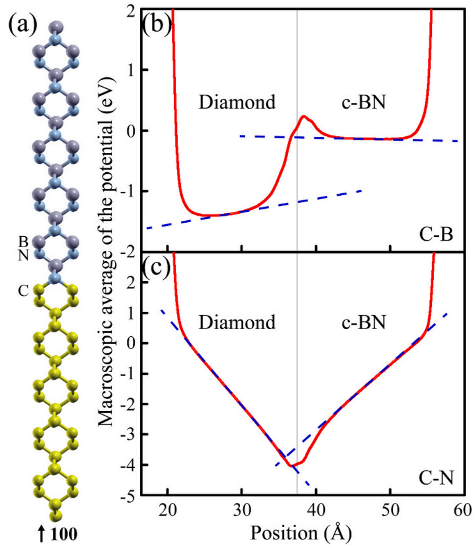

Fig. 1. (a) Atomic configuration of the diamond/c-BN (100) interface slab model used for interface charge density calculations, where only the symmetry-unique atoms within the primitive cell are shown. The C-B and C-N interface configurations are generated by exchanging the positions of B and N. (b, c) Macroscopic averages of the electrostatic potential for the C-B and C-N configurations at 300 K. The blue dashed lines represent linear fits to the bulk-like plateau regions.

Table 1
Interface charge density of the diamond/c-BN (100) heterostructure for the C-B and C-N interface configurations calculated under temperature-dependent lattice constants.

Temperature (K)	Lattice constant (Å)	C–B (q cm^{-2})	C–N (q cm^{-2})
300	3.572	-1.5×10^{12}	$+2.6 \times 10^{13}$
500	3.574	-1.6×10^{12}	$+2.6 \times 10^{13}$
700	3.576	-1.8×10^{12}	$+2.6 \times 10^{13}$
1000	3.580	-1.8×10^{12}	$+2.5 \times 10^{13}$

The listed lattice constants correspond to the temperatures indicated, as obtained from the thermal-expansion model. Here, q denotes the elementary electronic charge (1.6×10^{-19}C).

studies indicate that mixed C-B and C-N bonding configurations can arise under realistic synthesis conditions [35–37]. A simple area-weighted linear mixing estimate suggests that the net interfacial charge becomes positive once the fraction of C-N bonding exceeds approximately 5.5%. This result implies that diamond/c-BN heterostructures with mixed bonding configurations remain compatible with two-dimensional electron gas formation, providing a practical route toward thermally stable interfacial doping in realistic device structures.

4. Interfacial thermal resistance at diamond/c-BN interface

ITR, also referred to as Kapitza resistance [38] or thermal boundary resistance, arises from the mismatch of phonon transport properties at the interface between dissimilar materials. For epitaxial diamond/c-BN interfaces, phonon-mediated heat transport was evaluated using the phonon nonequilibrium Green's function (NEGF) formalism within the Landauer framework, as established in previous studies [39–44]. In this approach, phonon transmission across the interface is computed from first-principles interatomic force constants (IFCs), and the temperature-dependent ITR is obtained by integrating the phonon transmission over the Bose–Einstein distribution. Detailed derivations and numerical implementations of the phonon NEGF method are provided in Refs. 42–44 and are not repeated here.

ITR was evaluated for diamond/c-BN interfaces with (100) and (111) crystallographic orientations, which are commonly realized in heteroepitaxial growth. The (110) orientation was not considered due to its reported type-II band alignment, which is less relevant for interfacial charge accumulation at diamond/c-BN heterointerfaces [18]. Phonon properties required for the thermal transport calculations were obtained using density functional perturbation theory for periodic diamond/c-BN supercells (Fig. 2(a)). The calculations employed the same pseudomorphic constraint, exchange–correlation functional, and pseudopotentials as in the electronic structure calculations. An $8 \times 8 \times 8$ Monkhorst–Pack grid was used for phonon sampling.

Calculated phonon dispersions for bulk diamond and c-BN are shown in Fig. 2(b) and 2(c), respectively. Both materials exhibit broad phonon bandwidths and high-frequency longitudinal optical modes extending beyond 1200 cm^{-1}. The acoustic branches show comparable group velocities, indicating favorable spectral overlap in the low-frequency regime relevant for thermal transport. Phonon transmission spectra for the (100)- and (111)-oriented diamond/c-BN interfaces are presented in Fig. 2(d) and 2(f). The (111) interface exhibits higher transmission at low frequencies below approximately 200 cm^{-1}, while the (100) interface shows enhanced transmission over a broader frequency range, including higher-frequency modes.

The resulting temperature dependence of the ITR is summarized in Fig. 2(e) and 2(g). A crossover near 160 K is observed, below which the (111) interface exhibits lower ITR, whereas above this temperature the (100) interface shows consistently lower resistance. At 300 K, the calculated ITR values are 5.03×10^{-10} m^2 K W^{-1} for the (100) interface and 5.95×10^{-10} m^2 K W^{-1} for the (111) interface. With increasing temperature, the ITR decreases for both orientations, reaching 3.70×10^{-10} m^2 K W^{-1} and 4.83×10^{-10} m^2 K W^{-1} at 600 K for the (100) and (111) interfaces, respectively. The orientation-dependent ITR values obtained here provide quantitative parameters for subsequent electro-thermal device simulations.

5. Interfacial thermal resistance at diamond/a-Al_2O_3 interface

For diamond interfaces with a-Al_2O_3 gate dielectrics, direct application of atomistic phonon transport formalisms such as the phonon NEGF approach is not practical. a-Al_2O_3 lacks long-range periodicity, and realistic representation of structural disorder would require prohibitively large supercells. In addition, crystalline Al_2O_3 (e.g., sapphire) differs substantially in structure and vibrational properties from the a-Al_2O_3 films typically deposited by ALD in diamond FETs. As a result, crystalline Al_2O_3 cannot be used as a reliable proxy for the amorphous gate dielectric in first-principles phonon calculations.

In the absence of direct experimental measurements of thermal boundary resistance for diamond/a-Al_2O_3 interfaces, the diffuse mismatch model (DMM) [45] was employed to estimate ITR. Within the DMM framework, phonons are assumed to scatter diffusely at the interface, and the transmission probability depends only on macroscopic material properties. Under the gray approximation [46], the phonon transmissivity from material i to j is given by

$$T_{dij} = \frac{C_j v_j}{C_i v_i + C_j v_j}, \tag{5}$$

where C_i and v_i are the volumetric heat capacity and average phonon group velocity of material i, respectively. The corresponding ITR is expressed as

$$W = \frac{4[1 - 0.5(T_{d12} + T_{d21})]}{T_{d21} C_2 v_2}. \tag{6}$$

The elastic and thermal parameters used in the DMM calculations are listed in Table 2, including longitudinal sound velocity and volumetric heat capacity for diamond, Si, and a-Al_2O_3. For ALD-grown a-Al_2O_3, experimental studies employing picosecond acoustics report

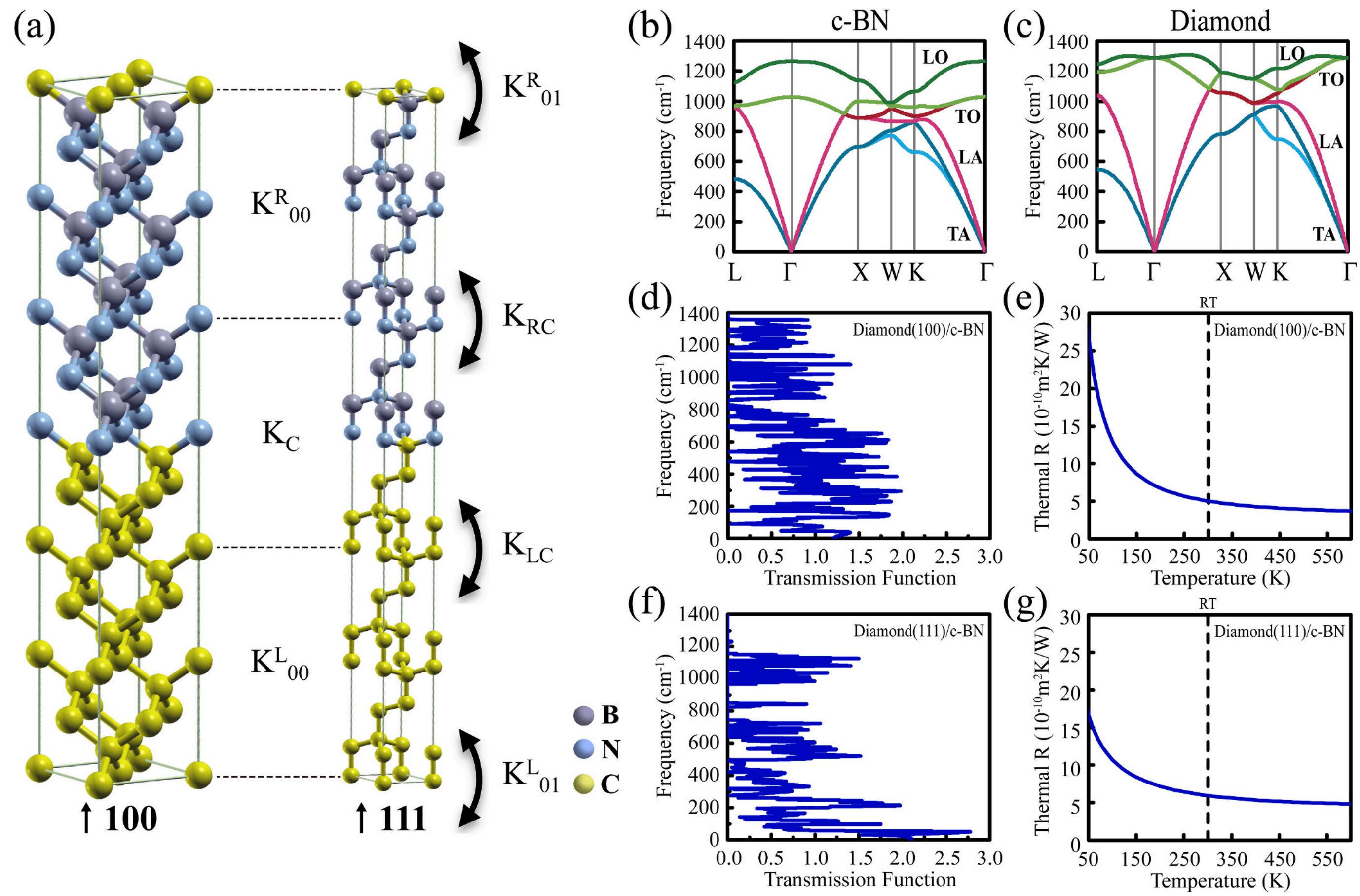

Fig. 2. (a) Atomic structures of the periodic lead–conductor–lead supercells used in interfacial thermal transport simulations, where the central regions represent the diamond/c-BN interfaces positioned between semi-infinite diamond and c-BN leads; the dynamical matrix K, which corresponds to the Hamiltonian in electronic transport, is constructed by IFCs [39–44]. (b, c) Calculated phonon dispersions of c-BN and diamond, respectively. (d, f) Phonon transmission spectra for the diamond/c-BN interfaces with (100) and (111) orientations, respectively. (e, g) Temperature-dependent ITR of the diamond/c-BN interfaces for the (100) and (111) orientations; RT denotes room temperature (300 K).

Table 2
Material properties adopted for interfacial thermal transport modeling and electrothermal device simulation.

Materials	Sound velocity ($m\ s^{-1}$)	Heat capacity ($10^6\ J\ m^{-3}\ K^{-1}$)	Thermal conductivity ($W\ m^{-1}\ K^{-1}$)	Bandgap (eV)	Dielectric constant	Critical electric field ($10^8\ V\ m^{-1}$)
Diamond	17500[45]	1.83[53]	2000[55]	5.47	5.70	10
c-BN	–	2.10[54]	1300[56]	6.40	7.10	2–6[57]
a-Al_2O_3	8400–9100[47–49]	2.35–2.93[49–51]	2.10[47]	8.80	9.30	2–8[58,59]
Si	8970[45]	2.07	–	–	–	–

longitudinal sound velocities in the range of approximately 8400–9100 $m\ s^{-1}$ at room temperature, despite variations in film density, growth conditions, and experimental methodologies [47–49]. Because transverse phonon modes are not well defined in amorphous materials, only the longitudinal component was considered, in accordance with the gray approximation. The volumetric heat capacity of a-Al_2O_3 is evaluated using a room-temperature specific heat of 0.88 $J\ g^{-1}\ K^{-1}$ and experimentally reported mass densities ranging from 2.67 to 3.33 $g\ cm^{-3}$ [49–51], resulting in a volumetric heat capacity range of approximately $2.35–2.93 \times 10^6\ J\ m^{-3}\ K^{-1}$.

Using the experimentally measured ITR of $6.92 \times 10^{-9}\ m^2\ K\ W^{-1}$ for Si/a-Al_2O_3 obtained by time-domain thermoreflectance (TDTR) [52] together with the DMM-based scaling, the diamond/a-Al_2O_3 ITR is estimated to be $(5.19–5.40) \times 10^{-9}\ m^2\ K\ W^{-1}$. This calibration is based on gray DMM calculations using the literature-based parameter ranges described above, which yield ITR ranges of $(1.83–2.09) \times 10^{-10}\ m^2\ K\ W^{-1}$ for the Si/a-Al_2O_3 interface and $(1.37–1.64) \times 10^{-10}\ m^2\ K\ W^{-1}$ for the diamond/a-Al_2O_3 interface. These values correspond to a DMM-based scaling ratio of approximately 0.75–0.78 for the diamond/a-Al_2O_3 interface relative to Si/a-Al_2O_3.

This calibration is motivated by the goal of anchoring the DMM-based estimation to experimentally grounded reference values, rather than by a direct comparison between fundamentally different modeling approaches such as phonon NEGF and gray DMM. While the Si/a-Al_2O_3 and diamond/a-Al_2O_3 interfaces may differ in processing conditions and surface preparation, the present approach does not assume their strict equivalence. In the absence of direct TDTR data for the diamond/a-Al_2O_3 interface, the experimentally characterized Si/a-Al_2O_3 system is used as a benchmark to obtain an experiment-anchored, engineering-level estimate for diamond/a-Al_2O_3. When the calibrated diamond/a-Al_2O_3 ITR is varied within the considered range in subsequent electrothermal device simulations, the predicted high-field response changes only negligibly at ambient temperature ($\Delta T_{max} \approx 1$ K and $\Delta I_{DS} < 0.1\%$ at $V_{GS} = -10$ V, $V_{DS} = -50$ V).

6. Electrothermal device behavior with a-Al_2O_3 and c-BN gate dielectrics

Electrothermal device simulations were performed to assess the impact of gate dielectric choice on the temperature-dependent operation of diamond FETs. All simulations were carried out using the Silvaco Atlas TCAD framework, with device geometries and material parameters chosen to closely reproduce experimentally reported structures. Temperature-dependent material properties, including thermal conductivity, bandgap, dielectric constant, and critical electric field, were incorporated based on literature values compiled in Table 2 [45,47–51,53–59]. ITR values and interfacial charge densities obtained from the preceding sections were implemented as fixed interface boundary conditions.

Carrier transport was modeled using temperature- and field-dependent mobility models based on Caughey–Thomas and Canali formulations [60,61], with parameters calibrated to experimental device output characteristics. While the detailed functional forms are not repeated here, the models explicitly account for mobility degradation with increasing temperature and electric field, ensuring physically consistent electrothermal behavior under high-bias operation.

Self-heating effects were included by solving the lattice heat-flow equation with Joule heating as the source term. Thermal boundary conditions were selected to represent a packaged device environment: the bottom surface of the diamond substrate was fixed at the ambient temperature to emulate an ideal heat sink, the top surface of the Si_3N_4 overlayer was assigned blackbody radiation, and all other boundaries were treated as thermally insulating. ITRs at the gate dielectric interfaces were implemented as fixed thermal boundary layers.

7. Model calibration

As a baseline, the simulation framework was calibrated using a hydrogen-terminated single-crystal diamond FET with a 100-nm-thick ALD-grown Al_2O_3 gate dielectric reported in Ref. 62. The reference device exhibits a peak drain current density of approximately 0.5 A mm^{-1} at $V_{GS} = -10$ V and $V_{DS} = -50$ V. Device geometry and contact configurations were reproduced as closely as possible (Fig. 3(a)), and the corresponding geometrical parameters and calibrated mobility settings are summarized in Table 3. Mobility-related parameters were adjusted to match the saturation-region drain current, which is the primary operating regime for power-device applications. A representative diamond/a-Al_2O_3 ITR of 5.3×10^{-9} m^2 K W^{-1} was adopted, corresponding to the midpoint of the calibrated range.

Fig. 3(b) compares the simulated output characteristics with experimental data. Good agreement is obtained in the high-conduction regime, particularly at large negative gate bias, validating the electrothermal model and calibration strategy. Minor discrepancies near the subthreshold region are attributed to the combined effects of interfacial electrostatics and mobility modeling, but do not affect the conclusions drawn from high-bias operation.

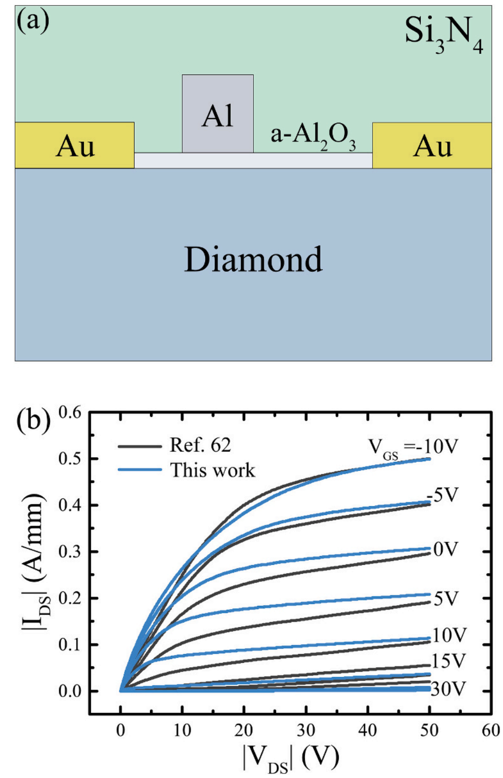

Fig. 3. (a) Cross-sectional schematic of the a-Al_2O_3-gated diamond FET used in electrothermal device simulations. (b) Output characteristics (I_{DS}–V_{DS}) for model calibration and validation: experimental data (black, Ref. 62) and simulated results (blue, this work).

Table 3
Key geometric and electrical parameters used in the modeled diamond FETs.

Parameters	Value
Gate length/thickness	0.9/0.5 μm
Source/Drain length	1.5 μm
Source/Drain thickness	0.3 μm
Source/Drain to gate gap	0.6/1.5 μm
Si_3N_4 thickness	1.0 μm
Gate dielectric thickness	0.1 μm
Substrate diamond thickness	300 μm
Source/Drain resistance	3.2 Ω mm
Interface defect trap density	9.0×10^{10} cm^{-2}
Interface charge density	-9.0×10^{12} cm^{-2}
T_μ	1.9
β_0	0.56
β_{exp}	1
v_{sat}	1.2×10^7 cm s^{-1}
μ	101 cm^2 V^{-1} s^{-1}

8. c-BN-gated diamond FET

The electrothermal response of a diamond FET employing a c-BN gate dielectric was evaluated using the calibrated simulation framework. To isolate dielectric-related effects within this framework, the baseline device geometry and transport settings were retained (Table 3), while dielectric-dependent parameters were updated, including material properties (Table 2), ITR, and interfacial sheet charge density.

Fig. 4(a) illustrates the simulated device structure for the c-BN gate stack under electron-channel operation. The source/drain and gate metals were set to Al ($\phi = 4.28$ eV) and Au ($\phi = 5.6$ eV), respectively. The diamond surface was assigned a positive electron affinity ($\chi = 4.3$ eV) to model ohmic electron injection at the contacts, guided by prior reports on nitrogen-terminated diamond surfaces [63–65]. The interfacial sheet charge density at the diamond/c-BN interface was set to 2.6×10^{13} cm^{-2} (C-N termination; upper-bound case), and the ITR was set to 5.03×10^{-10} m^2 K W^{-1} for a (100)-oriented interface.

Band offsets for the (100)-oriented diamond/c-BN interface with C-N bonding were adopted from Ref. 18 ($\Delta E_{CBO} = 0.59$ eV and $\Delta E_{VBO} = 0.34$ eV) and applied as boundary conditions for the device band diagram. To assess potential confinement at the interface, a one-dimensional Schrödinger–Poisson calculation was performed along the surface-normal direction (Fig. 4(b)). The calculated confinement was weak (first-to-second subband separation ≈ 52 meV; conduction-band edge to quasi-Fermi level separation ≈ 0.12 eV), and incorporating these quantized states produced negligible changes in the simulated I-V characteristics relative to drift–diffusion results. Accordingly,

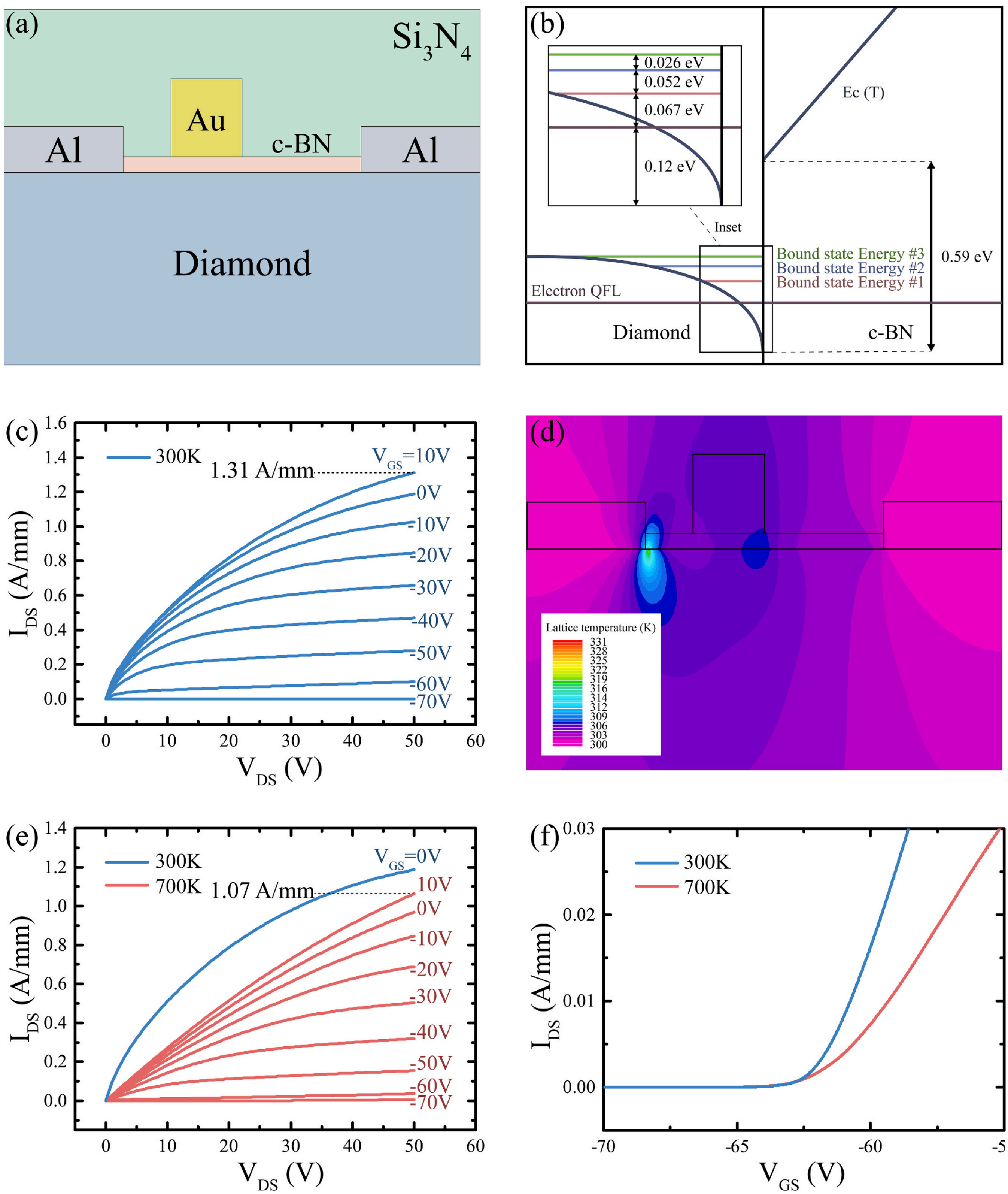

Fig. 4. (a) Cross-sectional schematic of diamond FET with a c-BN gate dielectric used in electrothermal simulations. (b) Conduction-band profile across the diamond/c-BN interface; the inset shows a magnified view of the quantized bound-state energies near the interface. (c) I_{DS}–V_{DS} characteristics at an ambient temperature of 300 K for the indicated gate biases. (d) Lattice temperature profile at 300 K under V_{GS} = 10 V and V_{DS} = 50 V. (e) I_{DS}–V_{DS} curves at 700 K (red, V_{GS} swept as indicated) and 300 K (blue, only V_{GS} = 0 V shown for comparison). (f) Transfer characteristics (I_{DS}–V_{GS}) at 300 K and 700 K.

drift–diffusion transport was used for the electrothermal simulations presented below.

Fig. 4(c) presents the output characteristics at an ambient temperature of 300 K, with V_{DS} swept from 0 to +50 V and V_{GS} stepped from −70 V to +10 V in +10 V increments. Under V_{GS} = +10 V and V_{DS} = +50 V, the peak drain current density reaches 1.31 A mm^{-1} under the specified interfacial charge condition. Fig. 4(d) displays the lattice temperature distribution under the same bias at 300 K. Two hotspot regions are observed: one near the source-side channel region and another near the drain-side gate edge. The peak lattice temperature reaches 320 K while the secondary hotspot exhibits a lower temperature of 308 K.

High-temperature operation was assessed by comparing output and transfer characteristics at 300 K and 700 K. Fig. 4(e) shows the output characteristics at 700 K (full V_{GS} sweep) overlaid with a representative 300 K curve (V_{GS} = 0 V) for comparison. At 700 K, the peak drain current density reaches 1.07 A mm^{-1} at the highest bias condition, corresponding to an approximately 20% reduction relative to 300 K. A similar reduction is observed across the gate-bias range in the saturation regime. Fig. 4(f) examines the transfer characteristics at 300 K and 700 K; the curves exhibit minimal horizontal shift over the plotted bias range, indicating limited change in the gate-bias onset of conduction under the modeled conditions.

To clarify the physical origin of the two hotspots observed in Fig. 4(d), additional electrothermal analyses were performed for the c-BN-gated diamond FET. Fig. 5(a) presents the Joule heat power density together with the simulation mesh, confirming that the regions of maximum heat generation are not associated with localized mesh refinement at the sharp metal contact corners.

Figs. 5(b) and 5(c) show the spatial distributions of the total current density and the electric field magnitude, respectively. The local maxima in Joule heat power density in Fig. 5(a) occur where strong electric fields spatially overlap with channel current. These two heat-generation maxima correspond to the two hotspots in Fig. 4(d). The primary hotspot coincides with the source-side channel entrance, where the channel first forms near the source contact and the associated electrostatics give rise to a locally enhanced channel-direction electric field under high-field operation.

A secondary region of elevated heat generation is observed beneath the drain-side gate edge, where the termination of gate control results in a localized enhancement of the electric field. In this region, the channel current density is lower and more spatially distributed than near the source-side channel entrance, leading to a secondary hotspot of lower intensity.

9. a-Al_2O_3-gated diamond FET (Baseline)

For comparison, electrothermal behavior was evaluated for the calibrated a-Al_2O_3-gated diamond FET under hole-channel operation. Fig. 6(a) shows the lattice temperature distribution at 300 K ambient under V_{GS} = −10 V and V_{DS} = −50 V corresponding to a drain current density of approximately 0.5 A mm^{-1}. The peak lattice temperature is localized near the drain-side gate edge at the dielectric interface, reaching 331 K, where the strong electric field is confined.

Fig. 6(b) compares output characteristics at 300 K and 700 K to assess temperature dependence. For both temperatures, V_{DS} is swept from 0 to −50 V; at 700 K, V_{GS} is stepped from +30 V to −10 V in −5 V increments, and a representative 300 K curve (V_{GS} = −5 V) is overlaid for comparison. At 700 K, the saturation drain current is reduced by approximately 40% across the plotted gate-bias range.

A direct comparison of the simulated electrothermal behavior highlights clear quantitative differences between the two gate dielectrics under the modeled conditions. At an ambient temperature of 300 K, the a-Al_2O_3-gated diamond FET reaches a peak lattice temperature of 331 K at a drain current density of approximately 0.5 A mm^{-1}, whereas the c-BN-gated device reaches a peak lattice temperature of 320 K while

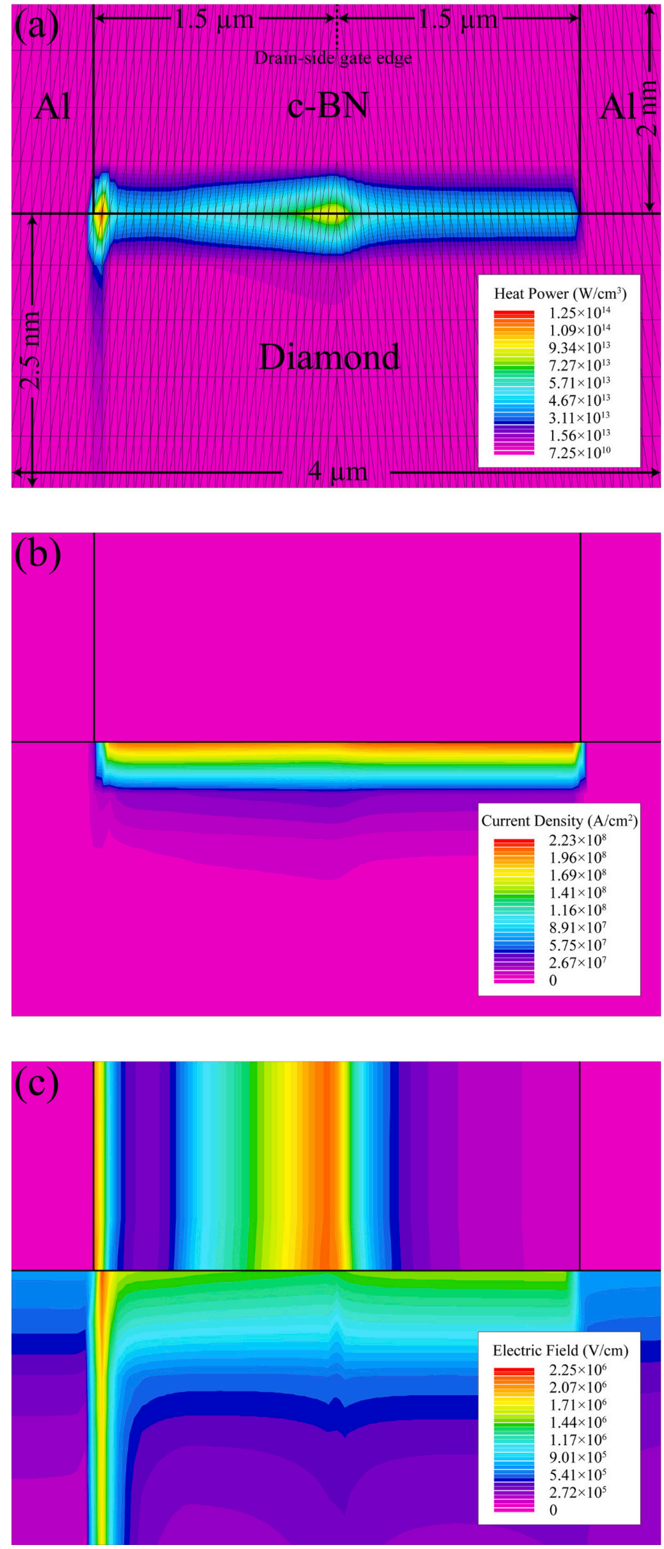

Fig. 5. Electrothermal analysis of the c-BN-gated diamond FET under high-field operation. All data correspond to the device biased at 300 K under V_{GS} = 10 V and V_{DS} = 50 V. The vertical dimension is not drawn to scale and is enlarged to provide a magnified view of the channel region. (a) Joule heat power density overlaid with the simulation mesh used in this work. (b) Spatial distribution of the total current density. (c) Spatial distribution of the electric field magnitude.

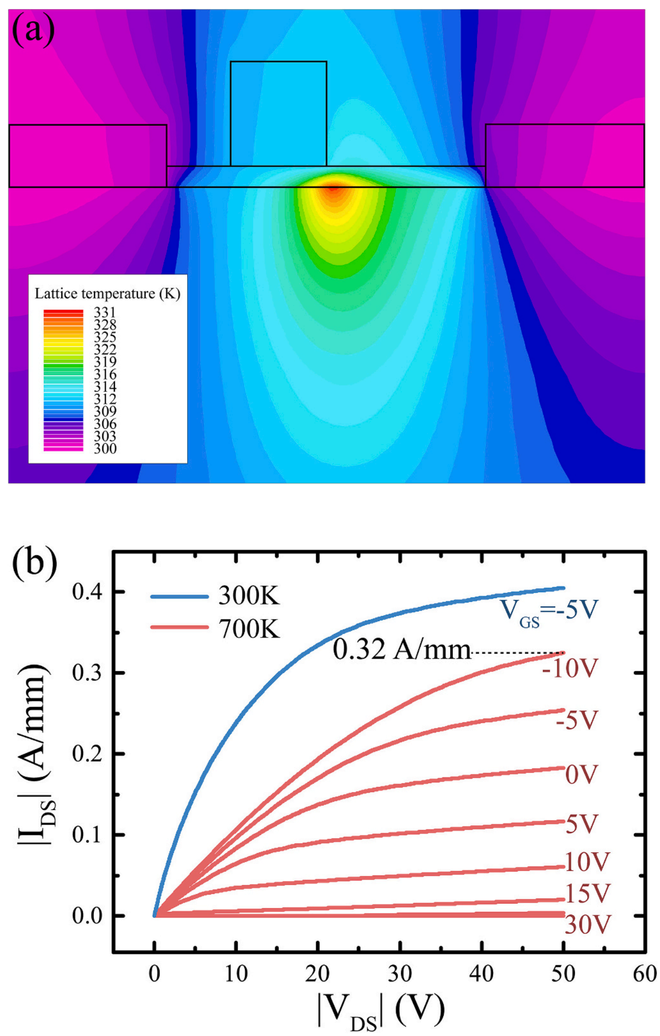

Fig. 6. Electrothermal response of the calibrated a-Al_2O_3-gated diamond FET (baseline) under hole-channel operation. (a) Lattice temperature distribution at 300 K ambient under $V_{GS} = -10$ V and $V_{DS} = -50$ V. (b) I_{DS}–V_{DS} characteristics at 700 K (red, V_{GS} swept as indicated) overlaid with a representative 300 K curve (blue, $V_{GS} = -5$ V) for comparison.

operating at a higher drain current density of 1.31 A mm^{-1}. Under elevated-temperature operation at 700 K, the a-Al_2O_3-gated device exhibits an approximately 40% reduction in saturated drain current relative to 300 K, while the corresponding reduction for the c-BN-gated device is approximately 20%. In addition, the c-BN-gated device shows minimal change in the gate-bias onset of conduction between 300 K and 700 K over the gate-bias range considered, whereas the a-Al_2O_3-gated device exhibits a more pronounced degradation in current drive. Together, these results underscore the distinct electrothermal responses associated with the two gate dielectrics within the present simulation framework.

For additional context, prior studies on highly scaled Si planar MOSFETs and FinFETs have reported temperature rises of ∼30 °C and up to ∼80 °C above ambient, respectively [66]. These values were obtained at current density levels comparable to those of the c-BN-gated diamond FET under $V_D = V_G = 1.5$ V. Considering that the c-BN-gated diamond FET exhibits a smaller temperature rise of ∼20 °C above ambient even at $V_{GS} = 10$ V and $V_{DS} = 50$ V, these results underscore the advantages of the c-BN-gated diamond FET for high-power operation under extreme bias conditions.

10. Conclusions

We developed a multiscale modeling framework that connects interfacial electrostatics and interfacial thermal transport to the electrothermal behavior of diamond FETs under elevated-temperature operation. By combining first-principles calculations with calibrated electrothermal device simulations, the framework enables quantitative evaluation of how gate dielectric choice governs temperature-dependent device performance.

Using a-Al_2O_3 and c-BN as representative gate dielectrics, we find that the diamond/c-BN interface supports polarization-induced interfacial charge that is weakly sensitive to lattice expansion up to 1000 K, while the diamond/a-Al_2O_3 interface does not provide an intrinsically temperature-stable charge mechanism. In parallel, phonon transport calculations show that the ITR of diamond/c-BN is nearly an order of magnitude lower than that estimated for diamond/a-Al_2O_3, reducing the thermal bottleneck at the gate dielectric interface.

When incorporated into electrothermal device simulations calibrated to experimental benchmarks, these interfacial differences lead to distinct temperature-dependent behavior. Under elevated-temperature operation, the a-Al_2O_3-gated device exhibits strong degradation in current drive, whereas the c-BN-gated device maintains higher current density, lower peak temperature rise, and minimal shift in the gate-bias onset of conduction within the modeled conditions.

These results indicate that, in diamond power devices, interfacial electrostatics and ITR are critical determinants of electrothermal performance once bulk thermal conductivity is sufficiently high. The framework presented here provides a quantitative basis for evaluating gate dielectrics and interface configurations in diamond FETs targeting high-temperature operation.

CRediT authorship contribution statement

Min Hyeok Kim: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Byoung Don Kong:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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